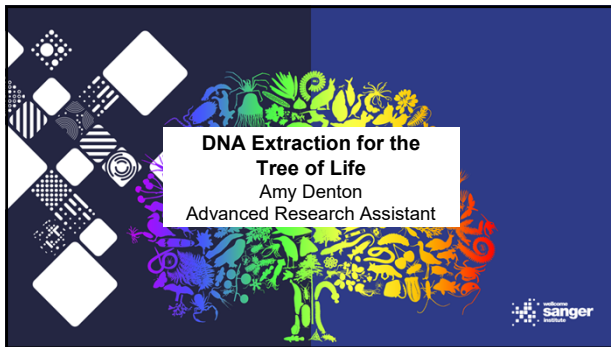




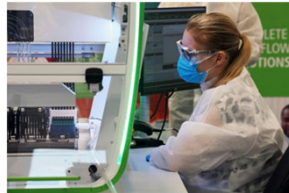
My career journey...

- Veterinary Medicine (with Zoology) Degree



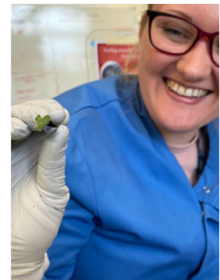
My career journey...

- Veterinary Medicine Degree
- Molecular Pathology Master's
- COVID laboratories



My career journey...

- Veterinary Medicine Degree
- Molecular Pathology Master's
- COVID laboratories
- Tree of Life

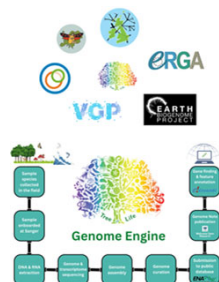


Aim of Tree of Life...

The overall goal of the Tree of Life Programme is to create reference genomes for various plant and animal species for several different projects.

This means sequencing the DNA from one individual of a species, assembling the pieces together, then finding the genes and other features.

The reference genome is then shared so other people can use this in further research - population studies, evolution, biology, conservation, biomedicines etc.



My job...

- Sample preparation
 - cryoPREP
 - powermashing
 - Bead beating
- DNA extractions
 - Manual
 - Automated
- PacBio Preparation (shear & spri)
- RNA extraction
- Quality control
 - QuBit
 - Femto
- R&D (research & development)



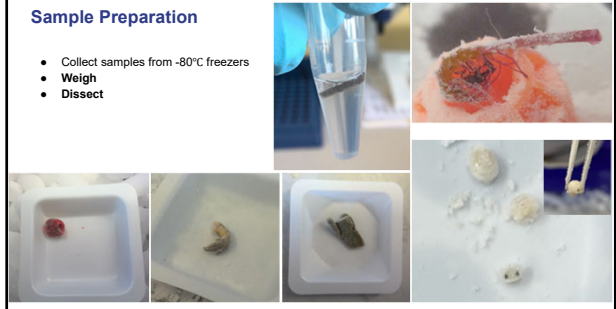
Sample Preparation

- Collect samples from -80°C freezers



Sample Preparation

- Collect samples from -80°C freezers
- Weigh
- Dissect



Sample Preparation

- Collect samples from -80°C freezers
- Weigh
- Dissect
- Disrupt via different methods



DNA Extraction

Different methods of DNA depending on the samples:

- MagAttract
- Plant Organic Extraction
- Nanobind
- Monarch
- Omega Bio-Tek

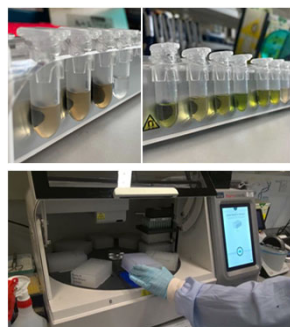


DNA Extraction

Different methods of DNA depending on the samples:

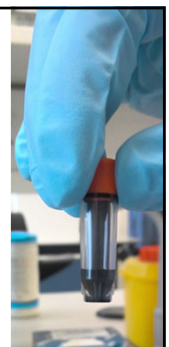
- MagAttract
- Plant Organic Extraction
- Nanobind
- Monarch
- Omega Bio-Tek

Extractions can also be manual or automated



Quality control

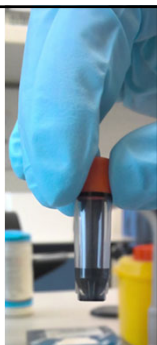
We need to know the quantity (how much) and the quality (how good) of the DNA we get from our animal and plant samples.



Quality control

We need to know the quantity (how much) and the quality (how good) of the DNA we get from our animal and plant samples.

QuBit - gives us an idea of how much DNA we have

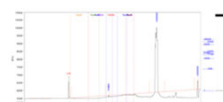


Quality control

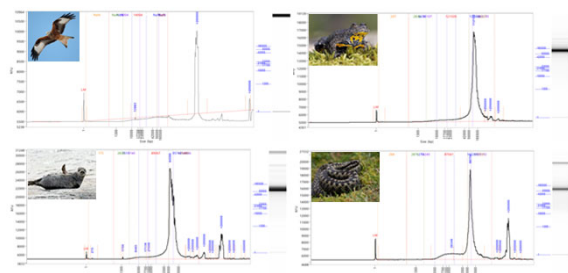
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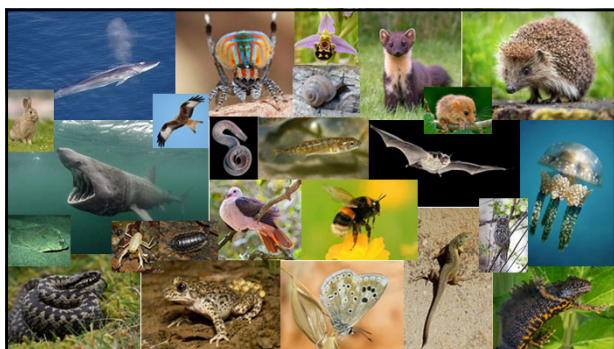
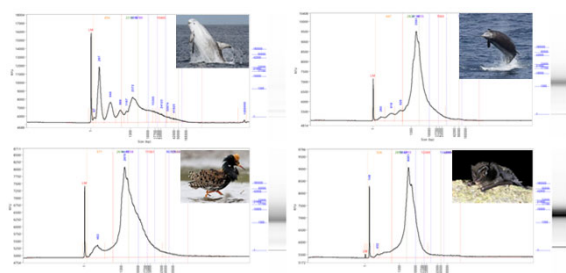
Femto - gives us an idea of how good the DNA is



Good quality DNA



Bad quality DNA





Genome quiz!

Which species do you think has the biggest genome?



A) Smooth Newt



B) Fin Whale



C) Buzzard

Genome quiz!

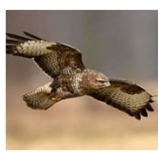
Which species do you think has the biggest genome?



A) Smooth Newt
25 Gb



B) Fin Whale
2.9 Gb



C) Buzzard
1.4 Gb

Genome quiz!

Which species do you think has the biggest genome?



A) Chicken of the woods



B) Mistletoe



C) Oak Tree

Genome quiz!

Which species do you think has the biggest genome?



A) Chicken of the woods
0.04 Gb



B) Mistletoe
87 Gb



C) Oak Tree
1 Gb

Genome quiz!

Which species do you think has the biggest genome?



A) Peacock Spider



B) Moon Jellyfish



C) Hagfish

Genome quiz!

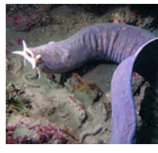
Which species do you think has the biggest genome?



A) Peacock Spider
8 Gb



B) Moon Jellyfish
0.6 Gb



C) Hagfish
2.7 Gb



Thank you for listening!

Any questions?

